Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Let $A = R - \{3\}, B = R - \{1\}$, and let $f : A \to B$ be defined by $\mathrm{f(x) = \frac{x-2}{x-3} fis}$

(A) not one-one

(B) not onto

(C) many-one and onto

(D) one-one and onto

Q.2

The image of the interval $[1, 3]$ under the mapping $f : R \to R$, given by $f(x) = 2x^3 - 24x + 107$ is

(A) $[0, 89]$

(B) $[75, 89]$

(C) $[0, 75]$

(D) None of these

Q.3

If $f : R \to R$ is given by $f(x) = \frac{x^2-4}{x^2+1}$, then the function f is

(A) many-one onto

(B) many-one into

(C) one-one into

(D) one-one onto

Q.4

Which of the following functions is (are) injective $\mathrm{map(s)?}$

(A) $\mathrm{f(x) = x^2 + 2, x \in (-\infty, \infty)}$

(B) $\mathrm{f(x) = |x + 2|, x \in [-2, \infty)}$

(C) $f(x) = (x - 4)(x - 5), x \in (-\infty, \infty)$

(D) $\mathrm{f(x) = \frac{4x^2+3x-5}{4+3x-5x^2}, x \in (-\infty, \infty)}$

@IITJEE_Advanced

Q.5

$f : R \to$ defined by $f(x) = (x-1)(x-2)(x-3)$ is

(A) one-one and into

(B) one-one and onto

(C) many one and into

(D) many-one and onto

Q.6

The number of bijective functions from a set A to itself when A contains 106 elements, is

(A) 106

(B) $(106)^2$

(C) (106)!

(D) 2^{106}

Q.7

$f(x) = \frac{\sin([x]\pi)}{x^2+x+1}$ where $[.]$ denotes the greatest integer function, then

(A) f is one-one

(B) f is not one-one and non-constant

(C) f is a constant function

(D) None of these

Q.8

Let function $f : R \to R$ be defined by
$f(x) = 2x + \sin x$ for $x \in R$, then f is

(A) one-one and onto

(B) one-one but NOT onto

(C) onto but NOT one-one

(D) neither one-one nor onto

Q.9

The signum function of: $\mathrm{R} \to \mathrm{R}$ given by

@IITJEE_Advanced

$f(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$

is

(A) One-one function

(B) Onto function

(C) One-One onto function

(D) None of these

Q.10

If the functions $f(x)$ and $g(x)$ are defined on $R \to R$ such that $f(x) = \begin{cases} 0, x \in \text{ rational} \\ x, x \in \text{ irrational} \end{cases}$ and $g(x) = \begin{cases} 0, x \in \text{ irrational} \\ x, x \in \text{ rational} \end{cases}$ then $(f-g)(x)$ is

(A) one-one and onto

(B) neither one-one nor onto

(C) one-one but not onto

(D) onto but not one-one

Q.11

Let $\mathrm{f} : \{\mathrm{x}, \mathrm{y}, \mathrm{z}\} \to \{1, 2, 3\}$ be a one-one mapping such that
only one of the following three statements is true and remaining two are false : $\mathrm{f(x)} \neq 2, \mathrm{f(y)} = 2, \mathrm{f(z)} \neq 1$, then

(A) $\mathrm{f(x)} > \mathrm{f(y)} > \mathrm{f(z)}$

(B) $\mathrm{f(x)} < \mathrm{f(y)} < \mathrm{f(z)}$

(C) $f(y) < f(x) < f(2)$

(D) $f(y) < f(z) < f(x)$

Q.12

If $f : R \to R, f(x) = \begin{cases} x|x| - 4, & x \in Q \\ x|x| - \sqrt{3} & x \notin Q \end{cases}$, then $f(x)$ is

(A) one to one and onto

(B) many to one and onto

(C) one to one and into

(D) many to one and into

@IITJEE_Advanced

Q.13

Let $f : R \to R$ be a function defined by $f(x) = \frac{x-m}{x-n}$ where $m \neq n$, then

(A) f is one-one onto

(B) f is one-one into

(C) f is many-one onto

(D) f is many-one into

Q.14

Which of the function defined below is one-one?

(A) $f : (0, \infty) \to R, f(x) = x^2 - 4x + 3$

(B) $\mathrm{f} : [0, \infty) \to \mathrm{R}, \mathrm{f}(\mathrm{x}) = \mathrm{x}^2 + 4\mathrm{x} - 5$

(C) $\mathrm{f} : \mathrm{R} \to \mathrm{R}, \mathrm{f}(\mathrm{x}) = \mathrm{e}^{\mathrm{x}} + \frac{1}{\mathrm{e}^{\mathrm{x}}}$

(D) $\mathrm{f} : \mathrm{R} \to \mathrm{R}, \mathrm{f}(\mathrm{x}) = \ell \mathrm{n}\left(\mathrm{x}^2 + \mathrm{x} + 1\right)$

Q.15

A function f from the set of natural numbers to integers defined by $\mathrm{f}(\mathrm{n}) = \begin{cases} \frac{\mathrm{n}-1}{2}, & \text{when n is odd} \\ -\frac{\mathrm{n}}{2}, & \text{when n is even} \end{cases}$ is

(A) neither one-one nor onto

(B) one-one but not onto

(C) onto but not one-one

(D) one-one and onto both

Answers & Solutions

Q.1 Answer:

one-one and onto

Solution:

Let $x_1, x_2 \in$ A and let $f(x_1) = f(x_2)$
or $\frac{x_1-2}{x_1-3} = \frac{x_2-2}{x_2-3}$
or $\mathrm{x}_1 = \mathrm{x}_2$So, f is one-one.
To find whether f is onto or not, first let us find the range of f.
Let $y = f(x) = \frac{x-2}{x-3}$ or $x = \frac{3y-2}{y-1}$
x is defined if $y \neq 1$, i.e., the range of f is $R - \{1\}$ which is also the co-domain of f. Also, for no value of y, x can be 3, i.e., if we put 3

@IITJEE_Advanced

$= x = \frac{3y-2}{y-1}$ **then** $3y - 3 = 3y - 2$ **or** $-3 = -2$
which is not possible. Hence, f **is onto.**

Q.2 Answer:

$[75, 89]$

Solution:

Since, the given function has minimum value 75 which is attained at $\mathrm{x} = 2$ **and maximum value 89 which is attained at** $x = 3$**. Hence, the range of** f **is** $[75, 89]$**.**

Q.3 Answer:

many-one into

Solution:

$\mathrm{f}(\mathrm{x}) = \mathrm{f}(-\mathrm{x})$**. So, f is many-one**
Also $\mathrm{f}(\mathrm{x}) = 1 - \frac{5}{\mathrm{x}^2+1} > 1 - 5 = -4$
So, f **is into**

Q.4 Answer:

$\mathrm{f}(\mathrm{x}) = |\mathrm{x} + 2|, \mathrm{x} \in [-2, \infty)$

Solution:

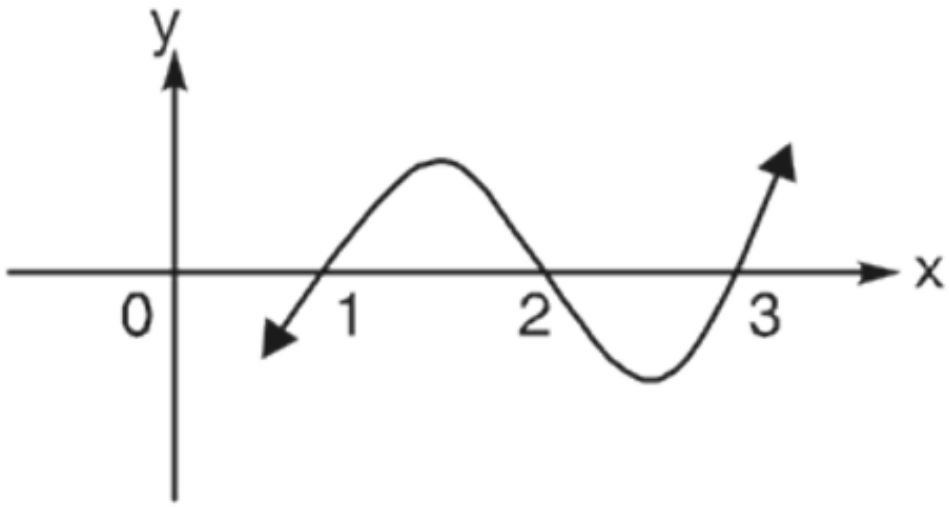
The function $\mathrm{f}(\mathrm{x}) = \mathrm{x}^2 + 2, \mathrm{x} \in (-\infty, \infty)$ **is not**
injective as $\mathrm{f}(1) = \mathrm{f}(-1)$ **but** $1 \neq -1$**. The function** $f(x) = (x - 4)(x - 5), x \in (-\infty, \infty)$
is not one-one as $f(4) = f(5)$**, but** $4 \neq 5$
The function, $f(x) = \frac{4x^2+3x-5}{4+3x-5x^2} x \in (-\infty, \infty)$
is also not injective as $\mathrm{f}(1) = \mathrm{f}(-1)$**, but** $1 \neq -1$**. For the function,** $f(x) = |x + 2|, x \in$
$[-2, \infty)$**. Let** $\mathrm{f}(\mathrm{x}) = \mathrm{f}(\mathrm{y}), \mathrm{x}, \mathrm{y} \in [-2, \infty) \Rightarrow |\mathrm{x} + 2|$
$= |y + 2|$
$\Rightarrow \quad x + 2 = y + 2 \Rightarrow x = y$
So, fis an injection.

Q.5 Answer:

many-one and onto

Solution:

@IITJEE_Advanced



Graphically, $y = f(x) = (x-1)(x-2)(x-3)$, which is clearly many-one and onto.

Q.6 Answer:

(106)!

Solution:

The total number of bijective functions from a set A containing 106 elements to itself is $(106)!$

Q.7 Answer:

f is a constant function

Solution:

$f(x) = \frac{\sin[x]\pi}{x^2+x+1}$ Let $[x] = n \in$ **integer**
$\therefore \sin[x]\pi = 0$ **or** $f(x) = 0$
Hence, $f(x)$ is constant function.

Q.8 Answer:

one-one and onto

Solution:

Given that $f(x) = 2x + \sin x, x \in R$
$\Rightarrow f'(x) = 2 + \cos x$
But $-1 \leq \cos x \leq 1$
$\Rightarrow 1 \leq 2 + \cos x \leq 3$
$\Rightarrow 1 \leq 2 + \cos x \leq 3$
$\therefore f'(x) > 0, \forall x \in R$
$\Rightarrow f(x)$ **is strictly increasing and hence one-one Also as** $x \to \infty, f(x) \to \infty$ **and** $x \to -\infty,$

@IITJEE_Advanced

$f(x) \to -\infty$

$\therefore$ **Range of** $f(x) = R =$ **domain of** $f(x) \Rightarrow f(x)$ **is onto. Thus,** $f(x)$ **is one-one and onto.**

Q.9 Answer:

None of these

Solution:

(a) $f(1) = f(2) = 1$

$\therefore I$ **and 2 have the same image**

$\Rightarrow \mathrm{f}\left(\mathrm{x}_1\right) = \mathrm{f}\left(\mathrm{x}_2\right) = 1$ **for** $\mathrm{x} > 0$

$\Rightarrow \mathrm{x}_1 \neq \mathrm{x}_2$

Similarly -1 **and** -2 **have the same image**

Here f **is not one-one**

(b) Except $-1, 0, 1$ **not other member of codomain of** f **has any pre image in its domain.**

$\therefore \mathrm{f}$ **is not on to**

$\Rightarrow \mathrm{f}$ **is not onto**

Q.10 Answer:

one-one and onto

Solution:

We are given that

$$f : R \to R, f(x) = \begin{cases} 0, x \in \text{ rational} \\ x, x \in \text{ irrational} \end{cases}$$

$$g : R \to R, g(x) = \begin{cases} 0, x \in \text{ irrrational} \\ x, x \in \text{ rational} \end{cases}$$

$\therefore (f - g) : R \to R \operatorname{sic} f - g)(x)$

$$= \begin{cases} -x \text{ if } x \in \text{ rational} \\ x \text{ if } x \in \text{ irrational} \end{cases}$$

Since $f - g : R \to R$ **for any** x **there is only one value of** $(f(x) - g(x))$ **whether** x **is rational or irrational.**

Moreover, as $x \in R, f(x) - g(x)$ **also belongs to** $\mathbf{R}$**. therefore,** $(f - g)$ **is one-one and onto.**

Q.11 Answer:

$\mathrm{f(x)} > \mathrm{f(y)} > \mathrm{f(z)}$

Solution:

Let $\mathrm{f(x)} \neq 2$ **be true and** $\mathrm{f(y)} = 2, \mathrm{f(z)} \neq 1$ **are false**

$\Rightarrow \quad f(x) \neq 2, f(y) \neq 2, f(z) = 1$

$\Rightarrow \mathrm{f(x)} = 3, \mathrm{f(y)} = 3, \mathrm{f(z)} = 1$**but then function is many one, similarly two other cases.**

@IITJEE_Advanced

Q.12 Answer:

many to one and into

Solution:

$f(2) = f\left(3^{1/4}\right) \Rightarrow$ **many to one function and** $f(x) \neq -\sqrt{3} \quad \forall x \in R \Rightarrow$ **into function**

Q.13 Answer:

f is one-one into

Solution:

Let $f : R \to R$ **be a function defined by**

$f(x) = \frac{x-m}{x-n}$

For any $(x, y) \in R$

Let $f(x) = f(y)$

$\Rightarrow \frac{x-m}{x-n} = \frac{y-m}{y-n} \Rightarrow x = y$

$\therefore$ **f is one** $-$ **one**

Let $\alpha \in R$ **such that** $f(x) = \alpha$

$\Rightarrow \alpha = \frac{x-m}{x-n}$

$\Rightarrow (x-n)\alpha = x - m$

$\Rightarrow x\alpha - n\alpha = x - m$

$\Rightarrow x\alpha - x = n\alpha - m$

$\Rightarrow x(\alpha - 1) = n\alpha - m$

$\Rightarrow x = \frac{n\alpha - m}{\alpha - 1}$ **· for** $\alpha = 1, x \notin R$

So, f **is not onto.**

Q.14 Answer:

$f : [0, \infty) \to R, f(x) = x^2 + 4x - 5$

Solution:

By definition only $f(x) = x^2 + 4x - 5$ **with domain** $[0, \infty)$ **is one to one.**

Q.15 Answer:

one-one and onto both

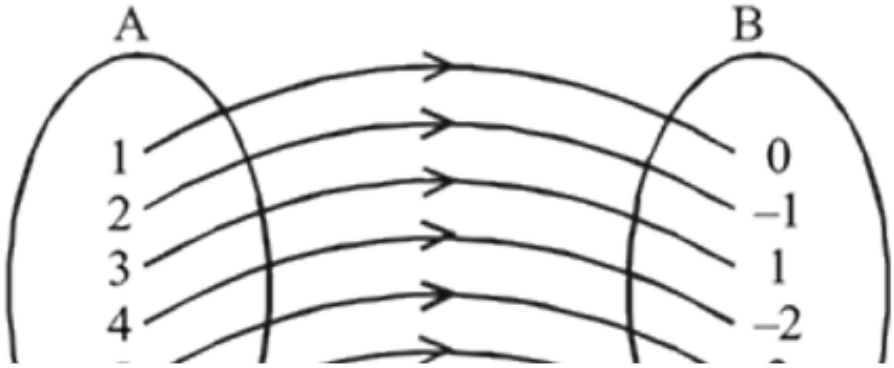
Solution:

$f : N \to I$

$f(1) = 0, f(2) = -1, f(3) = 1, f(4) = -2$

@IITJEE_Advanced

$f(5) = 2$, and $f(6) = -3$ so on



In thistype of function every element of set A has unique image in set B and there is no element left in set B. Hence f is one-one and onto function.

@IITJEE_Advanced